Final Report for Battery Management System (BMS)

Sponsored by Viking Motorsports (VMS)

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ECE Capstone
Senior Project Development

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1 Executive Summary

Our team will design the next-generation battery system for PSU's electric Formula race car, EV27, in partnership with Viking Motorsports and our faculty advisor. The current EV program does not have a usable accumulator, so the car won't run. Over the next two terms we will create a complete, competition-ready design for the high voltage battery pack and its Battery Management System (BMS), plus working low voltage prototype that demonstrates the core functions. The car we're designing this battery for will compete in a Formula-style racing competition where universities bring their own cars to do a race competition.

2 Background

VMS is our sponsor. They are a medium-sized student group that participates in the Formula SAE Electric event.

Electric vehicles rely entirely on large battery packs as their primary energy source. These battery packs are made up of many individual cells, typically lithium-ion, that must operate within some strict electrical and thermal limits and requirements in order for them to remain safe, efficient, and long-lasting. Due to the fact that lithium-ion battery chemistry is sensitive to properties such as overcharging, over-discharging, overheating/temperature, and cell imbalance, electric vehicles often require a dedicated subsystem called the BMS, to monitor and protect the battery.

These systems exist because lithium-ion battery packs can be extremely dangerous if not carefully monitored and diagnosed. The individual cells can be damaged due to things such as overcharging, over-discharging, and overheating. So, electric vehicles incorporate a battery management system to act as a protective mechanism using a combination of sensing, diagnostics, and protective circuit logic within the system. The system ensures that the battery operates only within safe electrical and thermal limits, which can in turn improve safety, extend battery life, and optimize performance.

The battery management system continuously measures key parameters such as cell voltages, current flow, pack and cell temperatures, and the estimated state of charge. It then processes the data to compute additional metrics, such as pack health, state of energy, imbalance levels, and thermal stability. In addition to monitoring, the BMS controls balancing circuitry using techniques such as active or passive cell balancing, to keep individual cells at consistent charge levels. Additionally, the BMS manages protective actions such as overvoltage, undervoltage, and overcurrent cut-offs. Also, the BMS communicates diagnostic information, warnings, and operational status to the vehicle's electronic control unit.

Within the larger electric vehicle system, the BMS serves as the central system responsible for determining whether the high-voltage battery pack is in a safe condition to deliver power to the vehicle. For instance, the electronic control unit depends on the BMS for accurate state of charge information, temperature status, and diagnostic reports. The motor, motor controller, charger, and thermal management system all rely on BMS outputs to function properly. Therefore, the BMS is not an isolated system but rather a core subsystem and safety component that interfaces with a lot of other components in the vehicle.

In practice, a battery management system is an electronic system that manages a rechargeable battery pack. Its operation follows a continuous loop of measure, protect, balance, and inform. First, it measures the fundamental state of the battery by using a network of sensors and integrated circuits to read the voltage of every individual cell, the total current flowing in or out of the pack, and its temperature at key locations. Second, it uses this data to protect the battery from damage by comparing the readings against safe limits. For instance, if it detects a dangerous condition like overvoltage, undervoltage, overcurrent, or extreme temperatures, it will command a safe shutdown by disconnecting the pack. Third, to maximize the pack's capacity and longevity, it balances the cells, using techniques such as active or passive cell balancing to bleed off excess charge from higher voltage cells, ensuring all cells reach a uniform state of charge. Finally, it informs the larger system by processing the raw measurements into useful information such as state of charge and health, and it communicates this data, along with any warnings, to other devices.

According to the project description provided by Viking Motorsports, the electrical team has completed early conceptual research for the Formula SAE Electric platform. This foundational work includes preliminary investigation into CAN bus communication for vehicle subsystem networking, initial evaluation of cell monitoring integrated circuits and measurement topologies for per-cell voltage sensing, and high-level planning of the system-level safety architecture required to meet Formula SAE Electric fail-safe shutdown and interlock rules. These efforts established an initial understanding of the technical challenges and constraints associated with a high-voltage accumulator system but did not result in an implemented or tested BMS design.

At the start of this capstone project, no functional BMS hardware or firmware prototype exists, and there is no previous generation of a Viking Motorsports BMS to revise or extend. As a result, this project represents a ground-up design and development effort rather than an incremental revision. Our team is responsible for transforming the sponsor's early conceptual research into a testable BMS prototype that performs real-time measurement, protection, balancing, and diagnostic functions that are suitable for future vehicle integration.

If nothing has been done on this project yet, are there comparable products or existing alternatives? Be creative here, what are similar technologies or direct competitors? Include them if they're relevant or help describe your project.

See commercial off the shelf products and open source projects sections of the project proposal.

2.1 Research

Final Summary: The BMS Knowledge Framework Matrix

This table represents the complete structure of our Proposal.

Layer	Module	Key Technologies / Keywords	Role
1. Sensing	**Voltage/Temp**	AFE Chips, NTC Thermistors, RC Filters	EE + ME
	Current	Hall Effect Sensor	EE + ME
2. Logic	**State Estimation**	SOC (Coulomb Counting), SOP (Derating)	CE
	Protection	OVP, UVP, OTP, OCP (Hard Limits)	CE
3. Acting	**Balancing**	**Passive Balancing** (Bleed Resistors)	EE
	HV Control	AIRs (Contactors), Pre-charge Circuit	EE + ME
4. Safety	**Isolation**	HV/LV Galvanic Isolation, Spacing	ME + EE
	Interlock	Safety Loop (Hardware series loop)	ME + EE
5. Comms	**External**	CAN Bus (Isolated), DBC File	CE
6. Env	**Integrity**	EMI/EMC, Shielded Cabling, Thermal Mgmt	ME + EE

2.1.1 Commercial Off-the-Shelf Products

Here are some options available to us for this project:

1. **Orion BMS 2:** The Orion BMS 2 is a widely adopted **Centralized** solution in the FSAE and EV conversion sectors. While it integrates all measurement

and control units into a single enclosure, offering mature reliability and EMI immunity, its architecture requires routing extensive sensing harnesses from all points of the battery pack to a central controller. This results in complex wiring and increased system weight. This serves as a critical counter-reference for our project: while centralized designs offer EMI advantages, their disadvantages in harness management within a high-voltage (300V+) and weight-critical racing environment validate our decision to pursue a Distributed Architecture.

2. **Stafl Systems BMS1000M:** Stafl Systems utilizes a **Master-Slave Distributed Topology**. The system consists of a master controller (BMS1000M) and multiple satellite monitoring modules (BMS1101S), utilizing differential signaling (e.g., IsoSPI) for communication. This architecture significantly reduces the length of analog sensing wires, effectively lowering wiring complexity within the high-voltage pack. As a direct industry benchmark, Stafl demonstrates the engineering feasibility of distributed architectures for segmented packs. However, its Closed-Source nature limits secondary development for specific research algorithms, restricting its role to a hardware architecture reference.

3. **EMUS G1 BMS:** EMUS G1 demonstrates a highly **Modular** distributed strategy, supporting the connection of multiple independent battery modules via CAN bus. This product employs robust galvanic isolation strategies and communication protocols ideal for physically separated battery enclosures (e.g., side-pod layouts). Given the potential requirement to split our 400V pack into multiple physical modules, EMUS provides a critical technical reference for High-Voltage Galvanic Isolation and data synchronization in our safety loop design.

2.1.2 Open Source Projects

Here are some projects we can borrow from:

1. foxBMS 2: foxBMS 2 is a comprehensive, professionally developed open-source BMS platform from the Fraunhofer research institute, explicitly designed as

a research and development platform for high-performance, high-availability battery systems. Its architecture is a master-slave distributed topology, where a central BMS-Master (ARM Cortex-R5) communicates via an interface board with daisy-chained Slave boards that handle cell monitoring (voltage/temperature) and passive balancing. The primary benefit is its exceptional depth. For instance, it offers a complete, well-documented hardware and software stack, including a built-in interlock line for safety shutdowns, CAN bus communication for system integration, and a flexible design that supports from a few cells up to several hundred kWh. The permissive licenses allow full use and modification. The notable drawback is its scale and complexity. For instance, as a generic research platform, it is not optimized for the strict space, weight, and FSAE-rule constraints of a race car. Its default use case is stationary storage, requiring significant adaptation for a mobile, high-vibration environment such as a race car. Nonetheless, it still serves as a great resource for robust system architecture, safety implementations such as interlock monitoring, and professional-grade, modular firmware structure.

2. Libre Solar BMS: The Libre Solar BMS is a compact, fully integrated open-source BMS designed for stationary and small-scale mobile DC energy systems (12V-48V, up to 16 cells). Its architecture is highly centralized, combining the BMS Analog Front-End (Texas Instruments BQ76952), microcontroller (ESP32-C3), power MOSFETs, and communication interfaces (CAN, UART, I2C, USB, Bluetooth) on a single PCB. The primary benefit is its production-ready, all-in-one design that demonstrates efficient integration of core BMS functions such as cell monitoring, protection, passive balancing, and contactor control, within a minimal footprint. Its firmware, built on the Zephyr RTOS, provides a professional reference for implementing safety logic, the ThingSet protocol for configuration, and state-of-charge estimation. The notable drawback is its scale. For instance, it is designed for 60V systems with currents up to 100A, its cell count and power handling are far below the requirements of a 400V+ FSAE powertrain. Its centralized architecture is also less ideal for the long, distributed battery pack typical in a race car frame. Nonetheless,

it still serves as an excellent reference for PCB-level integration, component selection, and a clean, modular firmware structure that can inform the design of sub-modules within our larger, distributed system.

3. Green BMS: Green BMS is a hobbyist-developed, smart BMS designed for small-scale lithium battery packs such as for electric scooters or solar storage. Its architecture is a modular distributed system, using simple Cell Modules based on ATtiny microcontrollers to measure individual cell voltage/temperature, which report via I2C to a central Control Unit (Arduino Mega). The system includes a separate "Limiter" unit for contactor control and an Android app for monitoring. The primary benefit is its exceptional transparency and simplicity, it demonstrates the absolute core functions of a BMS (monitoring, protection, balancing) with minimal complexity, using widely understood components (Arduino, ATtiny). This makes it an excellent educational tool for understanding the fundamental data flow and logic of a BMS before engaging with more complex, integrated circuits. The significant drawbacks are its lack of automotive robustness, missing critical safety features (high-voltage isolation, CAN bus, fail-safe hardware interlocks), and limited scalability (I2C communication is unsuitable for long daisy-chains in a large pack). Nonetheless, it still serves as a valuable conceptual reference for system partitioning and basic

2.1.3 Papers

Here is some extra papers we read during our research:

1. Evaluation and Comparison of Battery Cell Balancing Methods: This paper provides a structured, quantitative framework for selecting a cell balancing topology, directly addressing a key design decision. It categorizes and compares the three main balancing methods, dissipative (passive), capacitive energy transfer (active), and runtime balancing (active with converters), against critical metrics such as balancing speed, efficiency, complexity, and cost. This comparison is very valuable for our requirement to "support both active and passive cell balancing." It provides an objective basis to decide whether to

implement only passive (simpler, cheaper) or also include an active method (faster, more efficient) in our prototype.

2. Review of Battery Cell Balancing Methodologies for Optimizing Battery Pack Performance in Electric Vehicles: This paper provides a critical system-level perspective on cell balancing for batteries in electric vehicles, advocating for hybrid balancing methods that combine passive and active balancing techniques to leverage their respective advantages. This directly aligns with our requirement to "support both active and passive cell balancing," suggesting an approach where both techniques work together in a complimentary manner (passive for continuous maintenance, active for rapid pre-charge equalization). Additionally, it highlights environmental factors, specifically operating temperature, thermal uniformity, and mechanical vibration, as crucial but under-researched variables that affect long-term balancing performance and cell degradation.

3. State-of-the-Art and Energy Management System of Lithium-Ion Batteries in Electric Vehicle Applications: Issues and Recommendations: This paper provides a holistic overview of Li-ion battery technology and the complete scope of a Battery Management System (BMS). It systematically covers core BMS functions critical to our project, such as cell condition monitoring, state estimation (SOC/SOH), charge/discharge control, protection circuitry, cell equalization, and thermal management. It is particularly valuable for understanding the interdependencies between different BMS subsystems and for identifying key challenges, such as managing temperature gradients and diagnosing faults, which are actively relevant to the harsh, high-vibration environment of a Formula SAE vehicle.

3 Requirements

3.1 Product Overview

This paper provides a holistic overview of Li-ion battery technology and the complete scope of a BMS. It systematically covers core BMS functions critical to our project, such as cell condition monitoring, state estimation (SOC/SOH), charge/discharge control, protection circuitry, cell equalization, and thermal management. It is particularly valuable for understanding the interdependencies between different BMS subsystems and for identifying key challenges, such as managing temperature gradients and diagnosing faults, which are actively relevant to the harsh, high-vibration environment of a Formula SAE vehicle.

3.2 Stackholders

Our stakeholders include everyone who is directly or indirectly affected by the accumulator and BMS system. At the top level, we have our industry and university sponsors, who provide funding, facilities, and oversight, and care about safety, and reliability. The vehicle team are key users of our design: they integrate the pack into frame, rely on our interfaces, and depend on our documentation to build and service the system. Future student teams who will build, maintain, and possibly upgrade the pack are another important group, because our documentation will either make their work easier or harder.

3.3 Requirements

Must:

- Continuously monitor all cell voltages and pack temperature.
- Provide reliable overvoltage, undervoltage, and overcurrent protection.
- Support both active and passive cell balancing.

- Communicate with the Energy Control Unit (ECU) over the Control Area Network (CAN) bus.
- Operate within SAE EV voltage limits and isolation standards.
- Include a fail-safe shutdown or interlock function for critical faults.

Should:

- Include onboard data logging capability.
- Use modular architecture to scale for different battery pack sizes.
- Support self-diagnostic startup and fault recovery routines.

May:

- Include an integrated display or diagnostic interface for local readings.
- Support wireless telemetry or USB debugging for testing.

4 Objectives and Deliverables

4.1 Objectives

4.2 Deliverables

Using the Atlassian suite of tools which include Jira, Bitbucket, and Confluence to deliver on the following

- Project proposal
- Weekly Progress Reports
- Final report
- ECE Capstone Poster Session poster
- Detailed design documentation, explaining what your design does, how your design works, and why you made the decisions you did
- Electrical CAD: Schematics and board layouts, including output files (Gerbers, PDFs, etc)
- Mechanical CAD: enclosures, mechanisms, including output files (STLs, PDFs, etc)
- Bill of materials and pricing
- Developer's manual for future teams improveing on the existing design
- Short user manuals documenting how to use your creation
- A working prototype with a prototype battery pack and functionality with the ECU prototype.

5 Approach

6 Design

7 Test Plan Overview

Our test plan consists of a top

We're going to generate certain voltage and current values that qualify as Overvoltage, Undervoltage, and Overcurrent and the system will set the signal for the Shutdown Circuit and report each specific fault to the ECU with which specific cell(s) had the fault.

8 Results

9 Post Mortem

10 Future Work

11 Project Resources

12 Conclusion

Acronyms

BMS Battery Management System

BMU Battery Management Unit

CAN Control Area Network

CAN-FD CAN Fast Data

ECU Energy Control Unit

HV High Voltage

I^{2}C Inter-Integrated Circuit

LED Light Emitting Diode

LV Low Voltage

USB Universal Serial Bus

VMS Viking Motorsports

A Tooling

B Developer's Manual

C User's Manual

D Test Plan

This document lays out how someone is able to ensure the BMS is fully operational.

D.1 Top Down Testing

D.1.1 Purpose

To test the entire system against its requirements by performing a top-down test.

D.1.2 System Equipment

Equipment Needed

- System Equipment
  - Battery Monitoring Module
  - Battery Management Unit (BMU)
  - Current SHUNT Module
  - 1.5 miliOhm SHUNT Resistor
  - 12V battery with Low Voltage (LV) system power bus
  - Jumper wires for Daisy chain and Inter-Integrated Circuit (I^2C) interfaces
- Test Equipment
  - STLINK-V3MINIE
  - CAN-to-Universal Serial Bus (USB) connector
  - Hair Dryer
  - Multimeter
  - Zephyr Development Environment and STM32CubePorgrammer
  - Power Supply

- Other Equipment
  - Two battery modules with removable cells
  - Two 24V halogen light bulbs
  - Power cable connecting High Voltage (HV) system
  - HV switches

D.1.3 Pre-Test Setup

1. Attach the montioring boards to each of the battery modules. Ensure the exposed terminals of the thermistors are **connected to *negative* terminal of the cell they're measuring**.
2. Connect the SHUNT resistor LV terminals and Pack connectors to the Current SHUNT boards IN and V_{BUS} connectors respectively.
3. Assemble the rest of the HV system with the SHUNT resistor HV terminals, halogen bulb loads and switches. Only have two swtiches active.
4. Connect the following modules and interfaces:
   - The daisy chain connectors of the monitoring boards to each other. Then connect the south connector (COMML) of the first and the north connector (COMMH) of the last monitoring board to the COMM connectors on the BMU.
   - The I^{2}C interface between the Current SHUNT module and the BMU.
   - The Power Lines and Alert signal to the Header next to the I^{2}C interface on the BMU.
   - The STATUS_LED on the BMU to a simple active high Light Emitting Diode (LED).
   - The 12 volt power supply and ground to the LV system power bus.
   - The CAN Fast Data (CAN) interface on the BMU to the CAN-to-USB connector.
   - The STLINK-V3MINIE with the debugging port of the BMU

D.1.4 Top Down Test Steps

1. Tap the PWR button on the BMU to activate the BMS
2. If the BMU is not per installed with Firmware follow these instructions
   (a) open a terminal on the machine with the Zephyr Development Enviorment and STM32CubeProgrammer
   (b) Use the Zephyr command

   ```
   west build -p always -b nucleo_c092rc
   [Battery-Management-System-Capstone-2026 file path]/BMS_Firmware
   ```

   (c) Use the Zephyr command `west flash` to flash the BMU with the zephyr.elf file produced from the BMS_Firmware directory.
3. Read from device file connected to by CAN-to-USB connector
4. Measure the voltages of each cell, record the highest and lowest voltage, calculate the pack voltage, and compare to BMS voltage measurements.
5. Measure the voltage across the SHUNT resistor, calculate current, and compare that the BMS pack current
6. Measure the voltage of the SHUNTDOWN_REQ which should be logic low or 0V, observe the LED connected to STATUS_LED that should be currently off, and look at the message on the CAN-FD with id 0x010 which should be 0x00.
7. Blow hair Dryer on battery modules. SHUNTDOWN_REQ should be logic high or 5V, the LED is on, and the message has a value of 0x81.
8. Reset the BMS by turnning off and on the BMU
9. Now remove a cell from a Battery Module. SHUNTDOWN_REQ should be 5V, the LED is on, and the message has a value of 0x84.

10. Attach the power supply to the empty cell terminals and reset the BMS. SHUNTDOWN_REQ should be 5V, the LED is on, and the message has a value of 0x88.

11. Return the normal cell to its place, reset the BMS, and add activate the third bulb switch. SHUNTDOWN_REQ should be 5V, the LED is on, and the message has a value of 0x82.

12. Switch off the third bulb, reset the BMS, and disconnect **both** the daisy chain connectors on the BMU. SHUNTDOWN_REQ should be 5V, the LED is on, and the message has a value of 0x90.

13. Reconnect **both** the daisy chain connectors, reset the BMS, and use STM32CubeProgrammer to stop the program. SHUNTDOWN_REQ should be 5V, the LED is on, and the message has a value of 0xA0.

14. Turn off the BMS by tapping the PWR button again

D.1.5 Post-Test Teardown

1. Remove LV power bus from BMU

2. Remove all the connections between the modules

3. Disassemble the HV system

4. Remove the Current SHUNT module from SHUNT resistor and the Montior from each battery module.

D.2 Bottom-Up Testing

Each table below shows the test for each board.

Test Author: Jose Ramirez						
	Test Case Name:	CAN-FD Bus Test	Test ID #:			01
	Description:	Tests CAN-FD transmission	Type:			Black Box
Tester Information						
	Name of Tester:		Date:			
	HW/SW Version:	1.0	Time:			
	Setup:	Connect the a programmed BMU with BMS_CAN_Message_Test to a CAN-to-USB Connector				
TEST	INPUTS	EXPECTED OUTPUTS	PASS	FAIL	N/A	Comments
1						
2						
3						
4						
	Overall test result:					